

Faculty of Computing



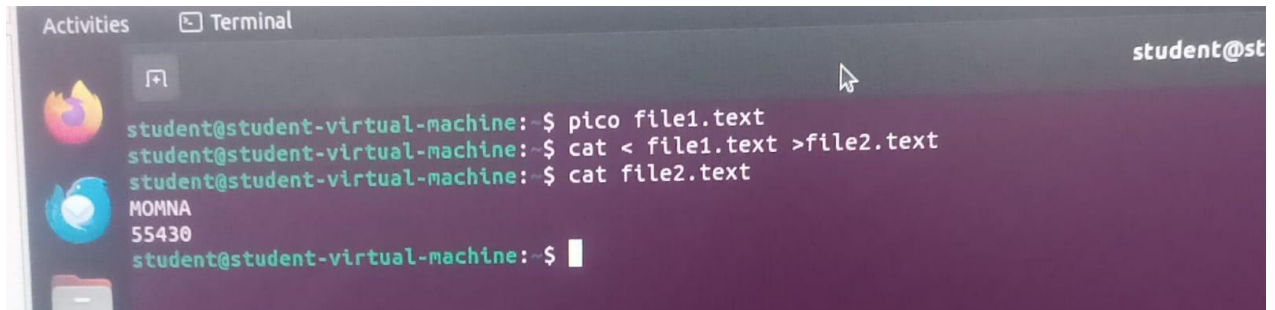
Operating Systems

Lab # 7

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Task 1

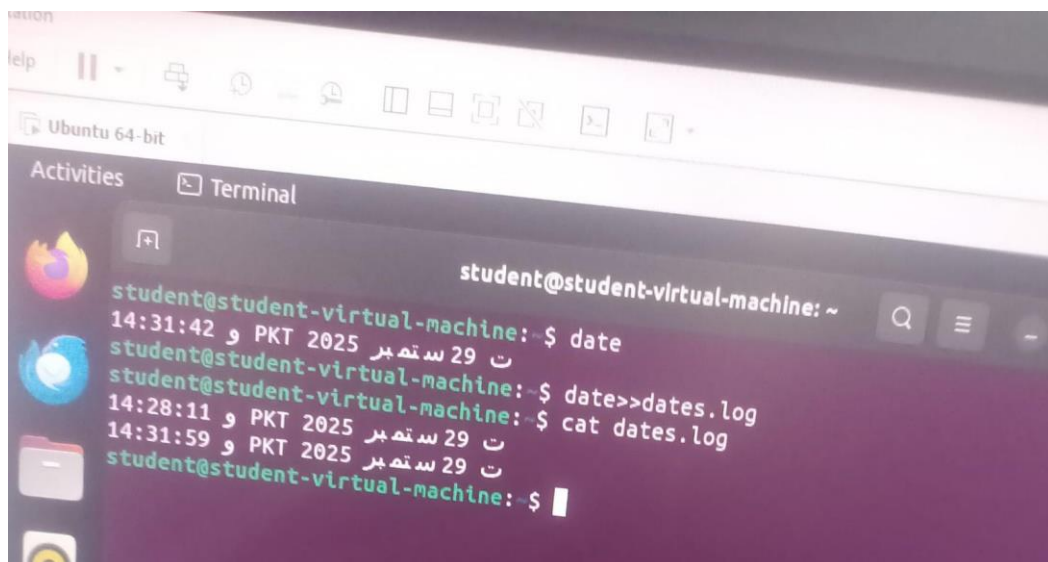
Read the contents of file1.txt and create file2.txt with the same content using redirection only in a single command.

A terminal window titled 'Terminal' showing a sequence of commands and their outputs. The user is logged in as 'student' on a 'student-virtual-machine'. The commands and outputs are: 'pico file1.txt' (no output), 'cat < file1.txt >file2.txt' (no output), and 'cat file2.txt' which outputs 'MOMNA' and '55430'.

```
student@student-virtual-machine:~$ pico file1.txt
student@student-virtual-machine:~$ cat < file1.txt >file2.txt
student@student-virtual-machine:~$ cat file2.txt
MOMNA
55430
student@student-virtual-machine:~$
```

Task 2

Run date and append the result to a file named dates.log every time you execute the command.

A terminal window titled 'Terminal' showing the execution of the 'date' command and its output being appended to a file. The user is logged in as 'student' on a 'student-virtual-machine'. The commands and outputs are: 'date' (output: '14:31:42 و PKT 2025 ت 29 ستمبر'), 'date>>dates.log' (no output), and 'cat dates.log' (output: '14:28:11 و PKT 2025 ت 29 ستمبر' and '14:31:59 و PKT 2025 ت 29 ستمبر').

```
student@student-virtual-machine:~$ date
14:31:42 و PKT 2025 ت 29 ستمبر
student@student-virtual-machine:~$ date>>dates.log
student@student-virtual-machine:~$ cat dates.log
14:28:11 و PKT 2025 ت 29 ستمبر
14:31:59 و PKT 2025 ت 29 ستمبر
student@student-virtual-machine:~$
```

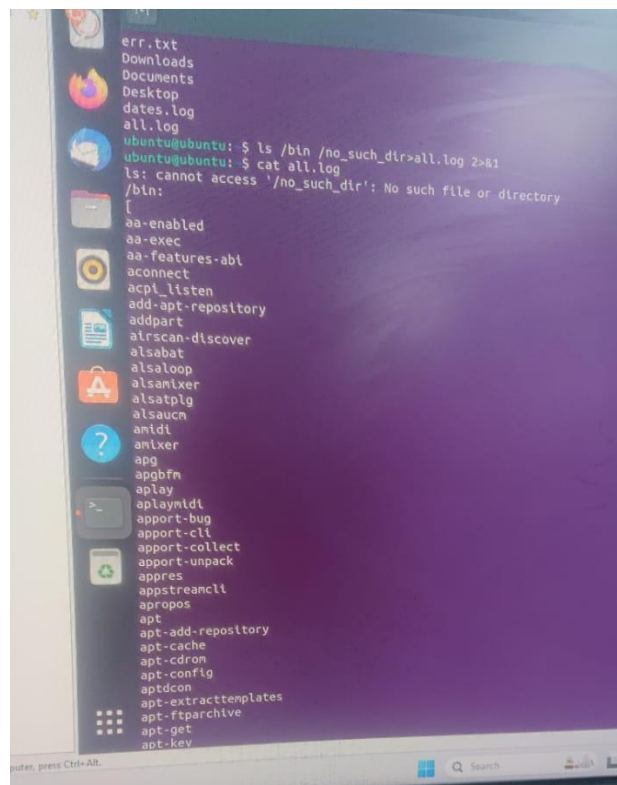
TASK 3

Run a command that will produce both valid output and an error. Redirect standard output to ok.txt and errors to err.txt.

```
student@student-virtual-machine:~$ ls /bin/no_dir 1 >ok.txt 2>err.txt
student@student-virtual-machine:~$ cat ok.txt
student@student-virtual-machine:~$ cat err.txt
ls: cannot access '/bin/no_dir': No such file or directory
ls: cannot access '1': No such file or directory
student@student-virtual-machine:~$
```

TASK 4

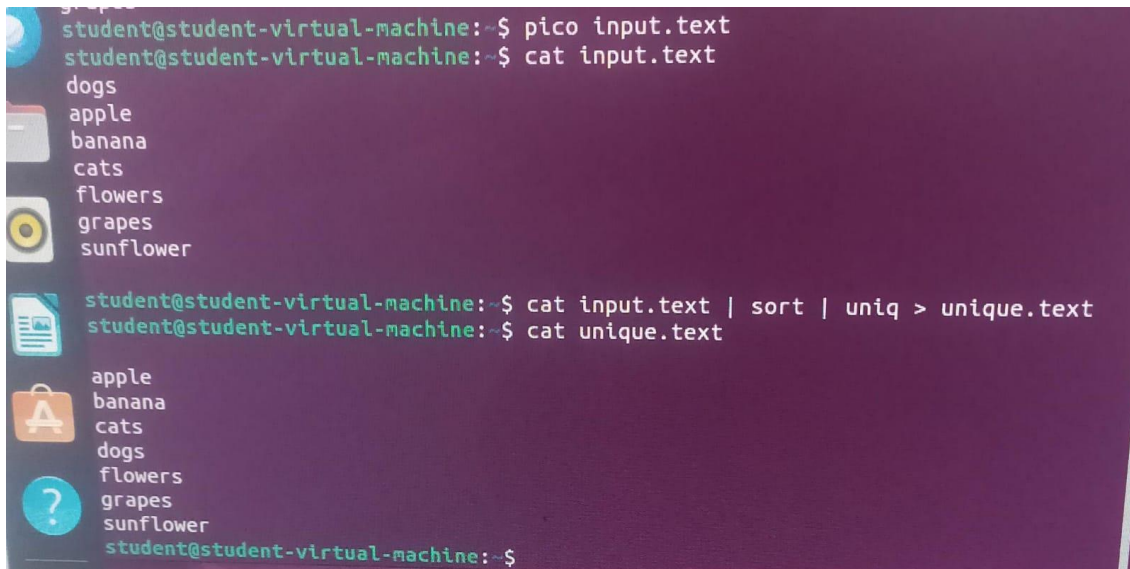
List the contents of the /bin directory and also try to list a non-existent directory /no_such_dir and redirect both the normal output and the error messages to be saved together in a single file called all.log in one line.



```
err.txt
Downloads
Documents
Desktop
dates.log
all.log
ubuntu@ubuntu:~$ ls /bin /no_such_dir >all.log 2>&1
ubuntu@ubuntu:~$ cat all.log
ls: cannot access '/no_such_dir': No such file or directory
/bin:
[
aa-enabled
aa-exec
aa-features-abt
aconnect
acpi_listen
add-apt-repository
addpart
aircan-discover
alsabat
alsaloop
alsamixer
alsatplg
alsaucm
amidi
amixer
apg
apgbfm
aplay
aplaymidi
apport-bug
apport-cli
apport-collect
apport-unpack
appres
appstreamcli
apropos
apt
apt-add-repository
apt-cache
apt-cdrom
apt-config
apticon
apt-extracttemplates
apt-ftparchive
apt-get
apt-key
```

TASK 5

Run a command that reads from input.txt and writes sorted, unique lines into unique.txt using pipe operator

A terminal window with a dark purple background and a sidebar on the left containing icons for a file manager, a terminal, a help icon, and an application store. The terminal shows the following commands and output:

```
student@student-virtual-machine:~$ pico input.txt
student@student-virtual-machine:~$ cat input.txt
dogs
apple
banana
cats
flowers
grapes
sunflower

student@student-virtual-machine:~$ cat input.txt | sort | uniq > unique.txt
student@student-virtual-machine:~$ cat unique.txt
apple
banana
cats
dogs
flowers
grapes
sunflower

student@student-virtual-machine:~$
```

TASK 6

What would be the result of the following commands.

- **cat filename > new** // filename is name of file that exist at current path
- **who > new**
- **ls | sort -r**
- **ls | sort -r >> new**

```
sunflower
student@student-virtual-machine:~$ pico filename
student@student-virtual-machine:~$ cat filename >new
student@student-virtual-machine:~$ who> new
student@student-virtual-machine:~$ ls |sort -r
Videos
unique.text
test_data.sh
Templates
taskt1.c
taskt1
taske.c
taskdd.c
taskdd
taskd.c
taskd
taskc.c
taskc
taskb.c
taskb
taska.c
taska
task5.c
task5
task4.sh
task4.c
task4
task3.sh
task3.c
task3
TASK2.sh
task2.sh
task2.c
task2
task1.sh
task1.c
task11.c
task11
task1
t5.c
t5
t4.c
t4
t3.c
t3
```

```
OK.txt
new
Music
labSort5
labSort
input.txt
filename
file2.txt
file1.txt
file
err.txt
Downloads
Documents
Desktop
dates.log
date.log
data.txt
Books
books
student@student-virtual-machine:~$ ls |sort -r>>new
student@student-virtual-machine:~$ cat new
student  tty2          2025-09-29 13:56 (tty2)
Videos
unique.text
test_data.sh
Templates
taskt1.c
taskt1
taske.c
taskdd.c
taskdd
taskd.c
taskd
taskc.c
taskc
taskb.c
taskb
taska.c
taska
task5.c
task5
```

mouse pointer outside or press Ctrl+Alt.

TASK 7

Sorts the contents of fruits.txt in alphabetical order. Filters the sorted output to only display fruits that contain the substring "ap"

```
student@student-virtual-machine:~$ pico fruit.txt
student@student-virtual-machine:~$ sort fruit.txt

Appale
apricort
banann
blueberry
grapes
orange
peach
pineapple
student@student-virtual-machine:~$ sort fruit.txt |grep "ap"
apricort
grapes
pineapple
student@student-virtual-machine:~$
```